

8. Hydrogen peroxide solution, H_2O_2 , is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover A	
Working temperature	= 50 °C
Cost per 100 cm ³	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm ³

Stain remover B	
Working temperature	= 30 °C
Cost per 100 cm ³	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm ³

Stain remover C	
Working temperature	= 20 °C
Cost per 100 cm ³	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm ³

Stain remover D	
Working temperature	= 30 °C
Cost per 100 cm ³	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm ³

(a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide

☐

type of oil used, towel material and temperature of stain remover

☐

type of oil used and towel material

☐

type of oil used, towel material and cost of stain remover

☐

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only

(b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover

☐

it is heat resistant

☐

it has a low concentration of hydrogen peroxide

☐

it takes a long time to work

☐

(c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

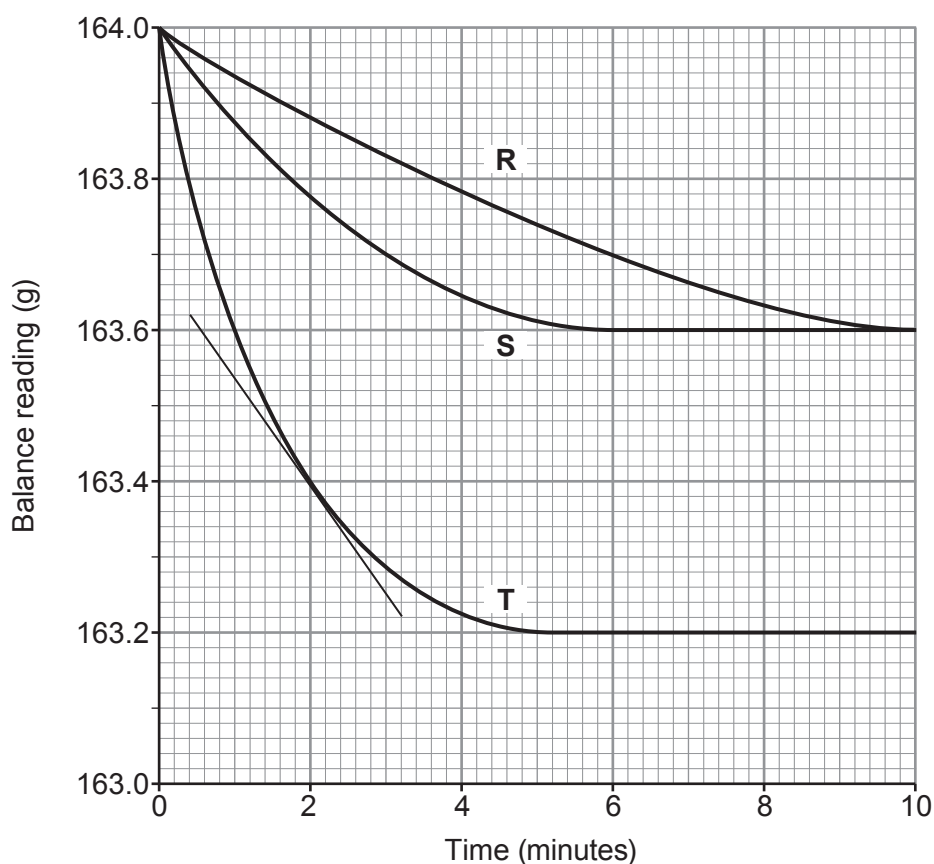


The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm³ of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



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- (i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = g/minute

- (ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

1	H	1
	Hydrogen	

7	Li	9	Be
	Lithium		Beryllium
3		4	
23	Na	24	Mg
	Sodium		Magnesium
11		12	
39	K	40	Ca
	Potassium		Calcium
19		20	
86	Rb	88	Sr
	Rubidium		Strontium
37		38	
133	Cs	137	Ba
	Caesium		Barium
55		56	
223	Fr	226	Ra
	Francium		Radium
87		88	
		227	Ac
			Actinium
		89	

11	B	12	C	14	N	16	O	19	F	20	Ne
	Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon
5		6		7		8		9		10	
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
	Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon
13		14		15		16		17		18	
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
31		32		33		34		35		36	
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Indium		Tin		Antimony		Tellurium		Iodine		Xenon
49		50		51		52		53		54	
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Thallium		Lead		Bismuth		Polonium		Astatine		Radon
81		82		83		84		85		86	

65	Zn	63.5	Cu	59	Ni	59	Co	56	Fe	55	Mn
	Zinc		Copper		Nickel		Cobalt		Iron		Manganese
30		29		28		27		26		25	
112	Cd	108	Ag	106	Pd	103	Rh	101	Ru	99	Tc
	Cadmium		Silver		Palladium		Rhodium		Ruthenium		Technetium
48		47		46		45		44		43	
201	Hg	197	Au	195	Pt	192	Ir	190	Os	186	Re
	Mercury		Gold		Platinum		Iridium		Osmium		Rhenium
80		79		78		77		76		75	
184	W	181	Ta	181	Ta	181	Ta	179	Hf	179	Hf
	Tungsten		Tantalum		Tantalum		Tantalum		Hafnium		Hafnium
74		73		72		72		72		72	
139	La	137	Ba	139	La	139	La	139	La	139	La
	Lanthanum		Barium		Lanthanum		Lanthanum		Lanthanum		Lanthanum
57		56		57		57		57		57	
227	Ac	226	Ra	227	Ac	227	Ac	227	Ac	227	Ac
	Actinium		Radium		Actinium		Actinium		Actinium		Actinium
89		88		89		89		89		89	

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number